

A BIM-based simulation framework for fire safety management and investigation of the critical factors affecting human evacuation performance

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ABSTRACT

Fire hazards are a big threat to human life and property safety. The U.S. fire statistics reveal that, in 2017 alone, 1,319,500 fires caused 3400 deaths and 14,670 injuries, which resulted in a loss of \$23 billion [1]. Effective evacuation planning in densely occupied buildings should be primarily put in place if both the number of injuries/fatalities and the level of property loss are to be minimized. However, it is not realistic, and is unethical to study human evacuation performance under a burning building. For this reason, computational tools tend to be the best approach for simulating fire growth as well as human response to fire hazards. This study aims to develop a BIM-based simulation framework that implements the Fire Dynamic Simulator (FDS) and agent-based modeling (ABM) for simulating fire growth and evacuation performance for different building layout scenarios. An experimental implementation is conducted to validate the proposed framework, which verified the benefits of (1) using BIM to offer a platform for conducting simulation design and visualizing the simulation results of (a) hazardous fire zones and (b) effective escape routes; (2) simulating fire growth using the FDS tool; (3) developing an agent-based model that accounts for the critical factors affecting evacuation performance; and (4) applying a statistical analysis for investigating the effects of influential parameters from the proposed model. As a result, the simulation outputs can be used to optimize the building design and to investigate the influential factors on human evacuation efficiency. The proposed framework contributes to building fire safety management by enabling to minimize both injuries/fatalities and property loss.

1. Introduction

The U.S. Fire Administration (USFA) has reported that, over the past decade, the fatalities and injuries, as well as the level of property loss due to fire hazards have increased. In 2017 alone, there were 3400 deaths, 14,670 injuries, and a loss of \$23 billion in property caused by 1,319,500 fires [1], which indicate an urgent need for improvement in current fire safety management practices. In addition, different from fires in other types of buildings, densely occupied buildings have a number of unique features, such as high-density pedestrian flows, rapid oxygen consumption, limited space for movement, and fixed exits. For this reason, effective evacuation planning in densely occupied buildings should be primarily put in place if both the number of injuries and fatalities and the level of property loss are to be minimized. Thus, appropriate building design is key to improving evacuation efficiency as well as maximizing the usable space within a building. However, it is unethical and not realistic to study human evacuation performance during a fire. Therefore, computational tools are widely accepted as the best approach for simulating fire growth and human response to fire

hazards. The simulation results can be used to predict fire safety performance and to assist in fire evacuation planning.

To date, advanced computational modeling concepts, such as building information modeling (BIM), Fire Dynamics Simulator (FDS), and agent-based modeling (ABM), have all been used to simulate the fire safety performance of buildings. Given the variety of modeling tools that are available on the market, it is arguably fortunate to take advantage of their respective strengths and implement them in fire safety management.

1.1. BIM in fire safety management

BIM is an advanced 3D modeling-based process that can be implemented to support the decision-making process with regard to a building or other built asset. Young et al. [2] reported that BIM have contributed to lean construction since mid-2000s. In fact, it has helped contractors to maximize project efficiency while also reducing the overall risks associated with a given project. As building fires are directly related to both casualties and building property losses, BIM

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implementation by design professionals have increased tremendously within the last decade due to the attempts to deliver better designs in terms of fire safety. Ruppel et al. [3], for example, designed a BIM-based serious game to determine human behavior during an emergency for evacuation planning. Similarly, Wang et al. [4] developed a BIM-based virtual environment to determine the evacuation flow given different building design options to optimize evacuation efficiency. In addition, Choi et al. [5] developed a BIM-based evaluation system to check the building regulations in place for fire safety assessment. Likewise, Eftekharirad et al. [6] integrated industry foundation classes (IFC) BIM and human behavior into a real-time fire emergency management system for fire safety control. Furthermore, Wang et al. [7] integrated BIM with a fire dynamics simulator to improve fire safety management practice, including evacuation assessment, egress planning, fire safety education, and fire equipment maintenance.

However, the full potential of such approaches has not yet been achieved. There are several obstacles including data interoperability as well as the technical limitations of the currently available BIM software, which cannot simultaneously model the fire-driven fluid flow and the occupant evacuation process. Therefore, this study aims to develop a comprehensive BIM-based simulation framework that is capable of implementing both FDS and ABM for fire safety management.

1.2. Advanced modeling concepts for BIM-based fire simulation

FDS is a computational fluid dynamics (CFD) modeling tool that is used to simulate fire-driven fluid flows. FDS software is interoperable with architectural BIM software, i.e. architectural BIM file can be input into FDS software to simulate fire growth and display the results through a visualization software known as Smokeview (SMV) [8]. To date, FDS has helped to achieve various fire simulation goals, such as post-accident investigations [9,10], fire safety assessments of existing buildings [11], and the analysis of factors that influence fire growth [12]. In an effort to integrate FDS with evacuation simulation tools (e.g., EVAC [13]), Tingyong et al. [14] developed a continuous model of FDS + EVAC to simulate the effects of fire on evacuation performance. However, they noted the difficulty of incorporating human behavior into an evacuation scenario design.

ABM technique enables to integrate human behavior into evacuation scenario design as it is a powerful modeling technique that is capable of facilitating both agent-to-agent interactions and agent-to-environment interactions. In ABM environment, agents can be defined as any type of individual that simulates behaviors in mathematical, theoretical, and logical ways [15]; it has been therefore applied across a wide range of research areas. For example, the prior applications of ABM in relation to evacuation scenarios have covered a variety of hazards, including earthquakes [16,17], tsunamis [18], wildfires [19], hurricanes [20], and volcanic activity [21] among others. Additionally, ABM can be applied during design, construction, or maintenance phases in order to assist in the decision-making process regarding a given building. Papadopoulos et al. [22], for example, integrated ABM with the building performance simulation (BPS) technique to simulate energy consumption and to deliver an energy system design. Similarly, Azar et al. [23,24] implemented ABM to investigate the effects of various occupancy parameters on energy usage in buildings during their maintenance phase. Therefore, it can be concluded that it is feasible to simulate indoor occupant responses using an ABM evacuation scenario design.

1.3. Studies relevant to indoor emergency evacuations

To date, a number of studies have been conducted with regard to the simulation of various emergency scenarios. Tang et al. [25] and Shi et al. [26], for example, designed ABM scenarios to simulate occupant performance in public building fires, while Peizhong et al. [27] designed an ABM simulation to investigate the evacuation outcomes in a

given subway station. In [28], Joo et al. developed an affordance-based model to simulate agent reactions to different exit routes. Additionally, Boguslawski et al. [29] proposed a dynamic approach using agent-based models to determine optimal exit routes in hazardous environments. Nevertheless, their studies implied the difficulties of simulating human response to disasters and predicting the evacuation performance.

In addition to the human behavior (both individual and social behavior patterns), evacuation performance depends on physical properties of buildings (e.g. building design (building layout and materials) and occupant capacity) as well as the fire characteristics (e.g. changes in temperature, toxicity, and smoke layer height) [30]. Therefore, in order to deliver moderate predictions of evacuation outcomes, the human response during an emergency, more specifically the critical factors affecting human evacuation performance, should be further investigated.

1.4. Objectives

The primary aim of this paper is to present a computational simulation framework for fire safety management. To that end, BIM serves as the platform for conducting the simulation and visualizing the simulation results—more specifically, simulating fire growth via the FDS tool and accounting for the critical factors in an ABM evacuation design. In order to determine its effectiveness, the proposed framework is tested on a set of data obtained from a night club fire. The simulation outputs are used to evaluate the fire resistance level and to predict evacuation outcomes for the studied building. Additionally, in terms of the statistical analysis of the simulation outputs, a linear regression model is used to optimize the building design; as well as two-sample t-tests are conducted to evaluate the proposed framework.

The remainder of the paper is organized as follows. Section 2 provides a comprehensive review of the relevant literature. Section 3 details the simulation framework and methodology. The experimental implementation and the results are then evaluated and discussed in Section 4. The final section draws conclusions based on the results and offers recommendations for future research.

2. Research background

A comprehensive review of the previous literature relevant to the proposed simulation techniques is provided in this section. Moreover, the critical factors that affect the evacuation time during building fires are also identified.

2.1. Building a collaborative BIM-based framework

In the past decades, several studies have implemented BIM for fire safety management [3–7]. Revit [31] is a BIM software that has been widely used in building design and construction due to the following properties. First, it is interoperable with a variety of simulation software, i.e. enables the import/export of common file formats to support the exchange of information [32]. Wang et al. [4], Choi et al. [5], Eftekharirad et al. [6], and Wang et al. [7], for example, have all successfully integrated data they obtained from other sources into Revit in order to design and simulate building emergencies. Second, the perspective and orthographic 3D views in Revit allow users to better visualize and more effectively communicate with the models. Third, Revit enables building design analysis by providing not only 3D information, but also property records [33]. For instance, Wang et al. [7] used Revit models to record fire safety equipment maintenance, hazardous areas, and planned escape routes. Admittedly, BIM environment enables contributions from multiple disciplines to a centrally accessed model, thus eliminates rework.

2.2. The FDS technique for fire growth simulation

PyroSim [34] software is used for simulating fire growth and was designed to act as a graphical user interface (GUI) that can effectively work with FDS models. The property libraries (e.g., material thermal properties, pyrolysis reactions, detector devices) provided by PyroSim outline the model parameters, which enhances the software's user-friendliness. Moreover, PyroSim enables the exchange of architectural information between FDS and BIM by enabling the transfer of models as DWG files [7,35–39]. During the transfer process, building materials and building compartments, i.e. building zones, are automatically grouped and made ready for users to customize their thermal properties. In addition, PyroSim writes post-processing simulation outputs that improve the efficiency of the simulation. More specifically, it allows users to visualize the fire dynamics as a 3D animation or 2D time history plots at any point during the analysis [35]. This enables users to be able to frequently access outputs to check, review, and even terminate the simulation scenario if the results are not satisfactory. Therefore, PyroSim is chosen as the FDS tool in this study as it enables to manipulate the fire simulation effectively and efficiently.

2.3. Evacuation simulation using ABM tools

2.3.1. Existing ABM software for evacuation scenario design

ABM tools have important applications in relation to building fire evacuation simulations for emergency safety management and construction safety control. NetLogo [40], Pathfinder [41], and AnyLogic [42] are all examples of currently available ABM software that supports the simulation of evacuation scenarios. Wagner et al. [43] and Liu et al. [44], for instance, built agent-based scenarios using NetLogo to foresee the evacuation outcomes during building fires. However, their models were restricted by the programming limitations during the model setup stage because solid programming language skill remains a prerequisite for using NetLogo. In addition, Mu et al. [45] implemented a FDS tool and Pathfinder software to analyze the evacuation time and egress options; however, their framework considered changing escape routes only, but did not look at optimizing the building space. Indeed, relevant modeling functions that can be simulated by Pathfinder effectively are limited. Alternatively, AnyLogic enables to import architectural BIM as a DXF file to exchange the building dimensional information [46]. Furthermore, it reflects agents' interactions and reactions to the defined scenario simultaneously. Consequently, among these software options, AnyLogic was chosen for this study due to its user friendly interface and ability to assess human evacuation performance on a BIM-based simulation platform.

2.3.2. Critical factors regarding human evacuation performance during building fires

As mentioned above, human evacuation behavior needs to be further investigated, while it remains significant to integrate the considered factors into the simulation design. After reviewing a wide variety of studies—including but not limited to [7,26,30,39,47–51], this study considers the three feature categories, namely human behavior, physical characteristics of the building, and fire conditions, that impact indoor fire evacuation performance, with each category consisting of a few critical factors (Table 1). In general, the efficiency of pedestrian evacuation performance depends not only on the decision-making process, but also on the speed of movement when evacuating. According to [52], an adult's average walking speed is 1.25 ± 0.3 m/s; however, that speed will start to be influenced by the smoke when the lower air layer reaches 1.5 m, and it will be further reduced when the lower air layer reaches 1.2 m [26,51] (Table 2). The framework presented here takes all of the listed critical factors that impact human evacuation performance into account for the accuracy of simulation outputs.

2.4. Investigating the effects of the critical factors on the evacuation time

Statistical analysis can help to investigate the simulation outputs rationally, thus validate and further evaluate a simulation system or framework. Galea et al. [10] and Wagner et al. [43], for example, analyzed their simulation outputs statistically to evaluate the effects of different building layouts during a fire scenario. Similarly, in [27], the simulation outputs were statistically analyzed to emphasize the simultaneous influence of fire conditions and human behavior on evacuation activities. Additionally, in [28] and [39], the outputs from different evacuation scenarios were applied in a linear regression model—a modeling approach in statistics—to investigate the linear relationship between the evacuation outcomes and various building properties (number of exits, width of the doorways, occupant capacity). Therefore, it can be concluded that statistical tools provide deeper insights into simulation outputs, which reinforce the reliability of the proposed works, which is why they are used in this study as well.

2.5. Contribution

This study proposes a comprehensive BIM-based simulation framework that primarily contributes to fire safety management by assessing the performance of fire resistance and evacuation safety in densely occupied buildings. In particular, architectural BIM bridges the technical gap between FDS and ABM for an efficient and effective simulation design. First, this framework reduces rework in terms of model setup as it enables import/export of FDS and ABM models during the modeling phase. Second, a wide range of research studies were reviewed to comprehensively summarize the critical factors with regard to the evacuation performance. Third, an evacuation scenario is proposed and validated through a representative case study—the Station nightclub fire, that was the fourth-deadliest nightclub fire in U.S. history. The results obtained from the case study indicated that the simulation outputs help to deliver moderate predictions of evacuation performance in building fires. Additionally, a statistical analysis was conducted and suggestion were made for fire safety strategies in terms of building layout design and evacuation planning. In conclusion, this study facilitates the use of advanced computational tools for fire safety management, which ultimately helps to minimize or avoid injuries/fatalities and property loss caused by building fires.

3. Research methodology

3.1. BIM-based simulation framework

The goal of this study is to develop a BIM-based simulation framework that is capable of implementing FDS and ABM to improve the accuracy of fire simulation results, thus fire safety management in densely occupied buildings. This should in turn help reduce fatalities and injuries as well as property loss during a fire hazard. The overall framework is shown in Fig. 1. The first step of the framework requires developing an architectural BIM for the facility of interest. In this study, we developed our architectural BIM, at Level of Detail (LOD) 300, using Autodesk Revit software. LOD 300, by definition, represents model elements graphically with precise geometry, meanwhile, non-graphical information, such as building material properties, is also attached to the model elements [53]. In this study, graphical information about building geometry is necessary for both fire simulation and evacuation simulation designs to represent building components in terms of quantity, size, shape, location, and orientation. Typically, non-graphical information does not need to be included in an evacuation scenario. However, non-graphical information, especially the information about building materials, is necessary for FDS analysis so that the thermal properties can be customized. Hence, an architectural BIM that meets at least LOD 300 should be used in the proposed framework. Nevertheless, it is important to note that an architectural LOD 300 BIM

Table 1
Critical factors affecting the fire evacuation performance.

Human behavior	Individual behavior	Risk and reward Bounded rationality Familiarity Proximity Task fixation	A psychological process of risk assessment for selecting escape routes before evacuating [47]. Bounded rationality prevents people from making rational decisions in emergencies [48]. Familiarity with the building layout drives people to choose the primary entrance as the exit [39,48]. People choose the closest exit to escape. People continue their movements toward the original escape route even it is proven to be ineffective [30].
	Social behavior	Herding behavior Crowd attraction Non-adaptive crowd behavior	Pedestrian movements are attracted to others' instructions [48]. Pedestrian movements are unconsciously attracted by crowds [48]. Destructive actions performed in crowd dynamics that cause a counter-flow effect [47–49].
Building properties	Building design	Alarm system Materials Exits Doorway width	The sensitivity of building alarm system may cause evacuation delays [47]. The thermal properties of materials heavily affect a building's fire resistance [30]. The number of available exits and their locations [39,44]. The capacity of exits to support pedestrian flow [39].
	Occupant capacity	Pedestrian flow density Visibility for way-finding	The pedestrian flow per unit area [30]. A crowd and chaotic room causes difficulties for way-finding [30].
Fire conditions	Fire growth	Temperature	The heat generated begins to harm the human body when (1) the upper layer radiation strength reaches 180 °C and (2) the layer of direct contact reaches 60 °C [7,50].
		Toxicity Smoke	The toxicity concentration of CO begins to harm the human body when it reaches 2500 ppm [7,50]. The smoke situation affects human moving velocity (1) as a 0.9 coefficient when the lower air layer reaches 1.5 m, (2) as a 0.6 coefficient when the lower layer reaches 1.2 m; and (3) it begins to hurt humans while the density reaches 85% [26,51].

Table 2
Pedestrian moving speeds and body radius.

Fire conditions	Pedestrian moving speeds (m/s)	Body radius (m)
1st ASET stage	1.25 ± 0.3	0.255 ± 0.035
2nd ASET stage	1.125 ± 0.27	
3rd ASET stage	0.75 ± 0.18	

may be readily available for a given facility. In that case, performing this first step would not be necessary. In this framework, the architectural BIM is exported as DWG and DXF files to be imported into FDS and ABM GUIs for conducting fire simulation and evacuation simulation, respectively.

The next step involves importing the architectural BIM into PyroSim for FDS analysis to test the building's fire resistance level that is represented by its available safe egress time (ASET) for pedestrians during a fire. PyroSim, a GUI that enables to build complex models for FDS, facilitates the exchange of architectural information (LOD 300) between BIM and FDS by importing LOD 300 BIM into PyroSim. During

the transfer process, the physical information of a building is seamlessly imported to represent the building geometry in the simulation space. Meanwhile, building materials and building components (i.e. building zones) are automatically grouped and made ready for users to customize their thermal properties for FDS analysis in PyroSim.

Likewise, the physical building information that is stored in the architectural BIM is also imported into AnyLogic, an agent-based modeling software environment, to simulate the evacuation scenario. During the import, AnyLogic obtains the building dimensional information (LOD 200) from the architectural BIM, which is exported as a DXF file from Revit, into the evacuation scenario to define the agents' movement area. The ABM evacuation scenario is designed based on the influential factors summarized in Table 1, and used to predict the required safe egress time (RSET) for occupants. It is important to note here that RSETs will change based on the number of exits, pathway widths, and occupant capacity of a facility. To achieve the intended fire safety performance, buildings are designed in a way that the RSET value is smaller than ASET value so that all the occupants have more than enough time (available time is more than required time) to safely egress the facility. The case where RSET and ASET values equal to each

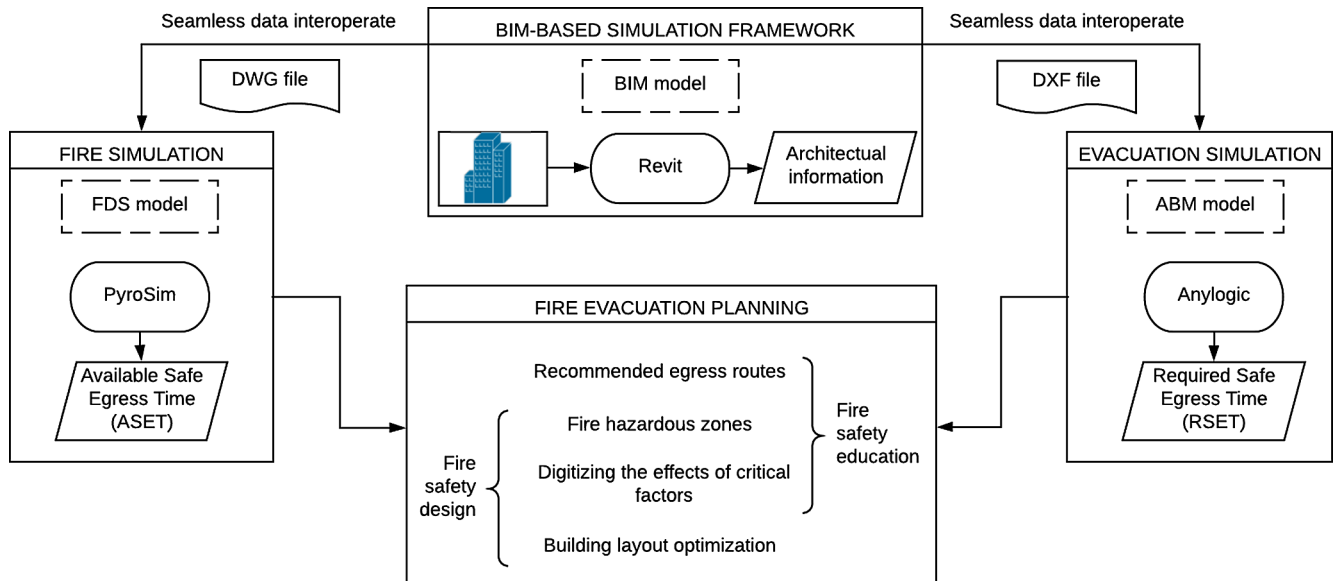


Fig. 1. BIM-based simulation framework.

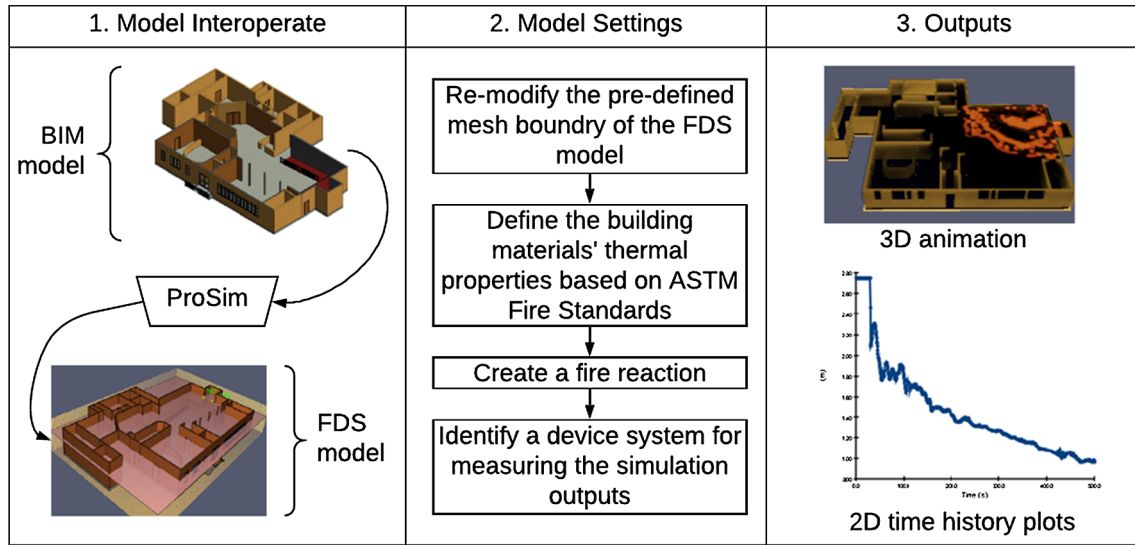


Fig. 2. Fire simulation module.

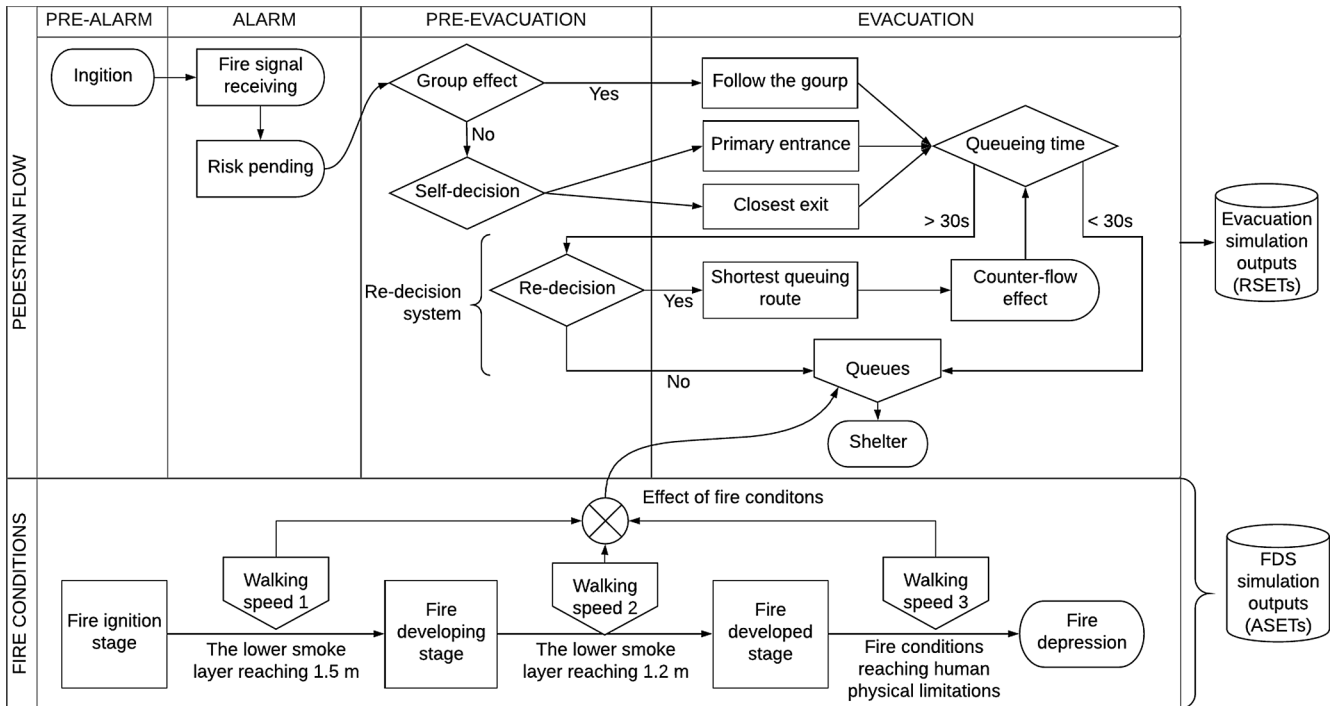


Fig. 3. Proposed evacuation simulation system.

other, nevertheless, could be considered as ideal since it ensures the maximum building space utilization while allowing time for all occupants to safely egress the facility, without casualties [52].

Next, the simulation results about hazardous zones (from FDS) and egress options (from ABM) are stored in a 3D-BIM environment and visualized using a 3D visualization software either on a desktop computer or a mobile device such as a tablet computer. In the final step, statistical analysis is conducted to evaluate the simulation outputs, thus to validate the proposed framework, and to optimize the building design for fire safety. In summary, the ultimate goal of this study is to minimize, possibly eliminate any fatalities and injuries and reduce the economic loss caused by building fires by implementing a 3D-BIM based framework for fire evacuation planning during the building design process, which can also be used for fire safety education.

3.2. Proposed simulation design

3.2.1. Fire dynamics simulation

The first step for fire dynamics simulation is to import the architectural BIM into PyroSim software, where it is resampled into a pre-defined 3D cubic mesh (Fig. 2). The next step involves setting model properties by remodifying the mesh boundaries to match it with the building geometry. It is noted that increasing mesh resolution results in more accurate simulation results, however, it also increases the computation time significantly. Once model mesh resolution is set, thermal properties and pyrolysis behaviors of building materials are meant to affect the fire reaction. Property libraries of PyroSim include the thermal properties of a limited number of sample materials. Thermal properties of other materials need to be retrieved from the ASTM Fire Standards and input into PyroSim model [54]. After defining the

material properties, a fire reaction is created by means of a burner surface with a specific heat release rate (HRR) and net heat flux (NHF), which are referred to as the power of the fire and the rate of heat energy transferred per surface unit area, respectively. Next, a device system is constructed to measure various pyrolysis products in terms of heat, gas, and fluid. As a result, dynamic measurements are presented as a 3D animation in the SMV program or as 2D time history plots in the statistic program embodied in PyroSim. The present simulation outputs of the temperature, toxicity, and smoke conditions are then used to assess the building's fire resistance level by determining its ASET for pedestrians.

3.2.2. Agent-based evacuation simulation system

The first step for evacuation simulation is to import the architectural BIM into AnyLogic to define the agents' movement areas. Fig. 3 illustrates two modules, namely the pedestrian flow module and the fire condition module, which form the evacuation simulation system as explained in detail below.

The pedestrian flow module is composed of the four periods of the evacuation process as following:

- The pre-alarm period indicates the simulation's starting point, which is defined by the fire ignition time.
- The alarm period measures an evacuation delay taking the time needed for risk recognition into consideration. The delay of pedestrian evacuation can be between 0 and 60 s depending on the sensitivity of the alarm system and the risk perception of individuals [30,47].
- The pre-evacuation period represents escape routes with agents that are driven either by social behavior (herding behavior and crowd attraction) or by individual behavior—specifically, evacuating toward the primary entrance due to its familiarity; otherwise, moving to the proximal exit while unaffected by bounded rationality or visibility for way-finding.
- The evacuation period represents agents' movements. It is assumed that a crowded doorway (i.e., the pedestrian flow exceeding 60 people/minute) require agents to queue to exit. With being affected by task fixation, agents keep their original escape routes even they are proven to be ineffective; conversely, a portion of agents may decide to choose an alternative route after having waited for longer than 30 s. In terms of counter-flow dynamics, non-adaptive behavior could reduce agents' movement speed by 19% [49]. Additionally, fire conditions would heavily affect the evacuation efficiencies as detailed below.

The fire condition module presents the changing speeds of agents' movements affected by fire dynamics in terms of the temperature, toxicity, and smoke conditions. These dynamic statistics can be linked to the FDS simulation outputs. In particular, the harmfulness of various fire dynamics to human body divide the ASET into three stages: (1) the smoke layer reaches 1.5 m (1st ASET stage); (2) the smoke layer reaches 1.2 m (2nd ASET stage); and (3) one of the pyrolysis products (heat generated, toxicity concentration, or smoke density) reaches human physical limitations (3rd ASET stage). Table 2 presents the moving speeds of agents at different ASET stages. The RSET, on the other hand, corresponds to the time required for all agents to exit the building uninjured/injured and reach a shelter.

During the simulation process, AnyLogic allows presenting agents' movements as a 2D/3D animation in real-time. The agents' behavior is defined as a simplified version of the AnyLogic-established model shown in Fig. 4.

3.3. Statistical analysis of designed systems

3.3.1. Linear regression model

Based on the simulation outputs, the next step is to conduct statistical analysis for fire safety management concerning building design

optimization and evacuation strategies. As mentioned above, in addition to human behavior and fire characteristics, indoor evacuation performance also depends on the physical properties of buildings such as building layout and occupant capacity. Specifically, adequate exit paths and appropriate occupant density can help to avoid crowding in certain building locations, e.g. near main building entrance, which could lead to evacuation failures. However, redundant exit paths would take extra floor space and negatively impact building space utilization. Hence, to achieve both fire safety and optimal building design and construction goals, it is desirable to maximize building space utilization while allowing time for all occupants to safely egress the facility. Otherwise, the building design (number of exits, pathway widths, and occupant capacity) is adjusted to achieve the case in which the RSET value is smaller than the ASET value. To assess the evacuation performance addressed by the simulation scenario, this study uses RStudio [55] as a tool to model the relationship between RSET and different building designs by fitting the linear regression equation provided below:

$$RSET = aX + bY + cZ + d \quad (1)$$

where the variable X represents the number of exits, the variable Y represents the width of the doorways, and the variable Z represents the occupant capacity. The factors a , b , and c represent the coefficients for X , Y , and Z , respectively, and factor d represents the slope-intercept of the line. The variable values (X , Y , or Z) can be modified in the simulation scenario, which is developed in AnyLogic, to generate corresponding RSET values as one of the simulation outcomes. Eq. (1) is used to compute the coefficients (i.e. a , b , c , and d) in RStudio. As a result, the linear regression equation can then be used either to estimate or to predict the evacuation performance for different building designs.

3.3.2. Two-sample t -test

To assist in evacuation strategies, it is important to investigate the impact of critical factors on evacuation performance. Table 1 provides a summary of these critical factors, which are represented with model parameters during the scenario design phase (Section 3.2). A two-sample t -test is used to evaluate the model parameters in terms of their effects on the simulation outputs. In advance, each model scenario is run multiple times to extract their average means as unbiased test data for t -tests. The basic idea behind a two-sample t -test is to compare the modified model scenario (removing one test parameter while keeping all the other parameters fixed) to the original model scenario (all parameters included) to verify the alternative hypothesis—that there is a difference between the two model scenarios—and to extract the estimated value of the mean difference. The estimated mean difference of RSETs validates the effectiveness of the test parameter. Accordingly, the validation of the model parameters evaluates the effects of their relevant critical factors with regard to the fire evacuation outcomes. The estimated value of mean difference, thereby, is used to measure the impact of relevant critical factor(s) to help improve evacuation strategies. If the estimated value is higher for example, it means that the factor has a higher impact on the evacuation performance.

4. Implementation and validation

4.1. Experiment overview

To validate the practicality of the proposed framework, it is feasible to conduct an experimental implementation using a well-documented fire emergency that has (1) precise physical information of the building to develop an architectural BIM at LOD 300 and (2) adequate records of injuries/fatalities to evaluate the simulation outputs. The proposed framework supports 3D geometry, thus multi-story buildings can be evaluated using the proposed framework, which can give more insights for stairways as escape routes compared to the single-story case study presented in this paper. However, due to the limited access for well-

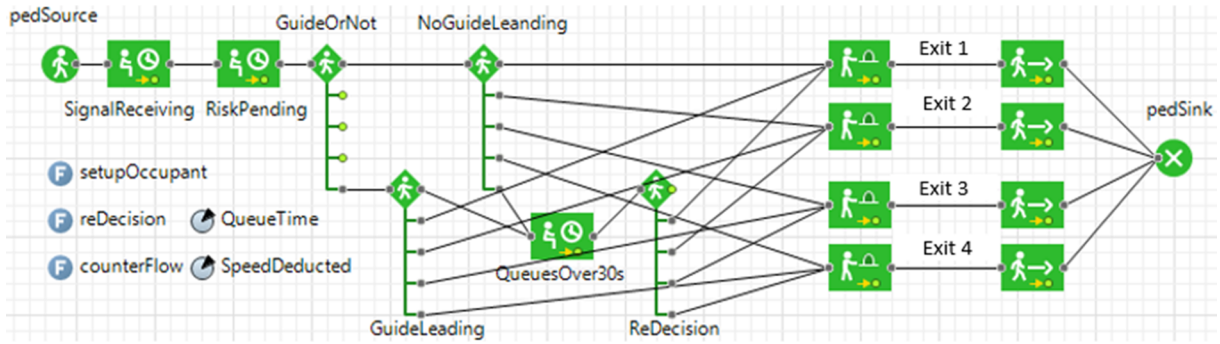


Fig. 4. A simplified version of agents' behavior in AnyLogic.

documented fire events, the Station Nightclub fire, which occurred in 2003, was finally chosen to be used as the proof-of-concept study in this paper. The chosen case is particularly significant since it was the fourth deadliest nightclub fire in U.S. history. To clarify, the fire killed 100 people and injured 230, with only 132 people escaping uninjured (from a total of 462 people present in the club that night). The accident report [56] indicates that: (1) the pyrotechnics ignited the non-fire-retardant polyurethane foam located in the alcove, which resulted in a fire; (2) the unprotected wood structure accounted for 95% of the fuel load, which caused harsh fire spread; and (3) a majority of evacuees jammed the primary entrance rather than identifying the three side exits, which affected evacuation efficiency negatively.

The first step for framework implementation is to develop an architectural BIM at LOD 300 (Fig. 5). However, we did not have access to the construction drawings of the building when establishing our architectural BIM. For this reason, we obtained and used the building dimensions included in the accident investigation report published by

NIST [56]. Likewise, the types of building materials were also obtained from the NIST report [56].

Prior to fire simulation step, the architectural BIM (LOD 300) was exported from Revit in DWG format and imported into PyroSim to conduct the fire simulation. In PyroSim, building materials and components are automatically grouped and made ready for simulation. The next step is to manually customize the thermal properties of materials and create a fire reaction in the simulation space. The thermal properties of major building materials were obtained from ASTM E84 [54] (Table 3). The effects of minor building materials, however, are negligible as they make a small portion of the fuel load. Further details about the fire simulation design can be found in Section 3.2.1.

At the end of the fire simulation step, the ASET value is computed using FDS analysis in PyroSim environment, which is then used to develop the fire condition module contained in the evacuation simulation scenario. The procedures for the evacuation simulation design are described in Section 3.2.2 in detail. The evacuation scenario design

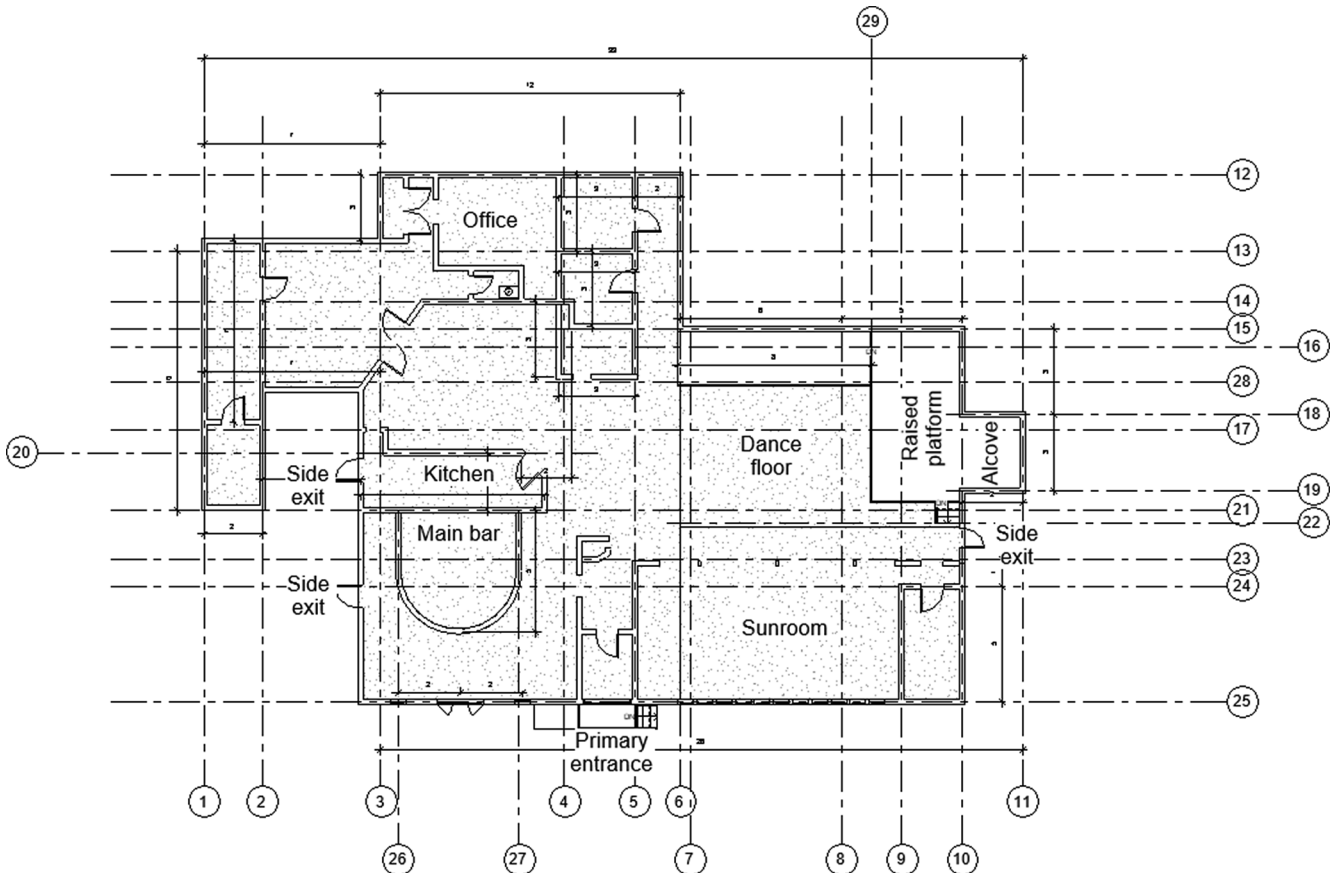


Fig. 5. The night club floor plan in Revit.

Table 3
Thermal properties of building materials.

Material Name	Density (kg/m ³)	Specific Heat (kJ/(kg·K))	Conductivity (W/(m·K))
Wood Panel	513	1.38	0.115
Concrete	1600	0.84	0.79
Nylon Carpet	128	1.42	0.1
Polyurethane Foam	22	1.4	0.034

starts with importing the physical building information that is stored in the architectural BIM into AnyLogic as a DXF file (LOD 200) in order to define the agents' movement area. The agents' movement process is defined in the flow chart presented in Fig. 4. The agents' parameters are modified using specific model parameters embodied in AnyLogic.

Next, by changing the value of model parameters, various scenarios were examined and the relevant experimental outcomes were computed for statistical analysis as described in Section 3.3. It is important to note here that the experiment assumes that: (1) each test scenario is independent of the others, (2) the FDS model defines exponential fire growth, and (3) agents with designed properties are randomly distributed throughout the evacuation scenario.

4.2. Fire simulation results

Fig. 6 illustrates different fire growth stages under both timelines of the fire accident and the SMV animation. As can be seen, the correspondence between the two timelines validates the accuracy of our fire simulation. In PyroSim, the time history plots generated by the statistic program indicate that: (1) the smoke layer reaches 1.5 m at 180 s, while it reaches 1.2 m at 300 s (Fig. 7a), which will cause a reduction in pedestrian movement speed of 10% and 40%, respectively; (2) the smoke density reaches 85% at 380 s (Fig. 7b), which occurs earlier than the other human physical limitations (i.e., tolerance to temperature and toxicity) in the simulation outputs. In addition, the SMV animation shows that the dance floor area accounts for the fastest fire spreads (Fig. 8). In summary, the ASET stages of this simulation can be determined as (1) 180 s for uninjured escape (1st ASET stage), (2) 300 s for injured escape without death (2nd ASET stage), and (3) 380 s for failure to escape alive (3rd ASET stage).

4.3. Evacuation simulation results

In the evacuation scenario, the number of agents in all three ASET stages indicates their injuries/causalities. Each scenario was run ten

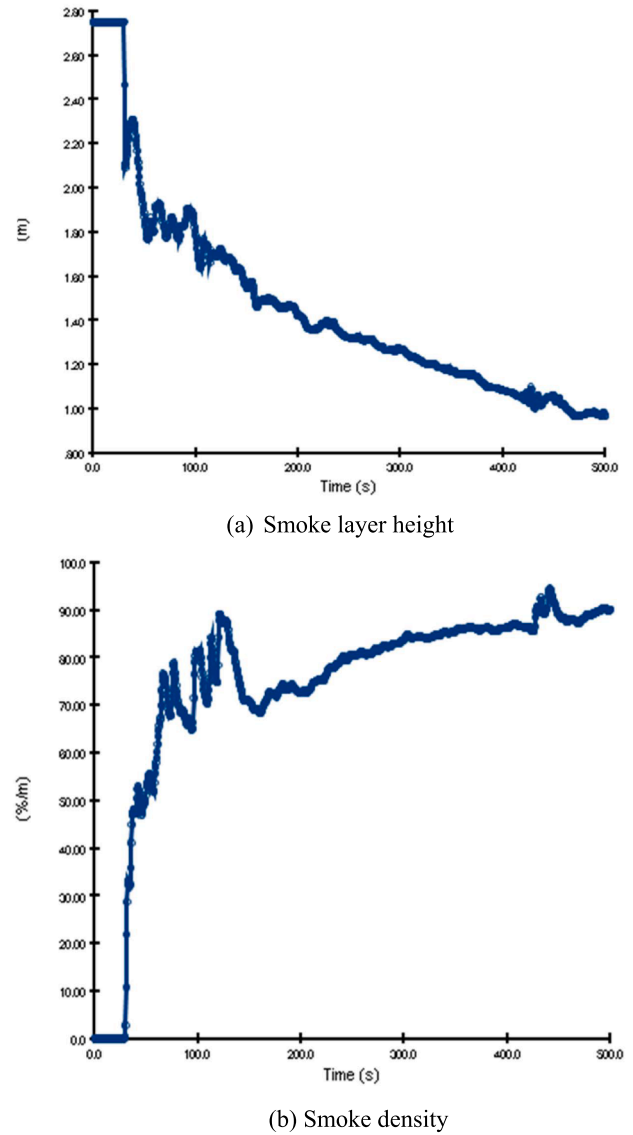


Fig. 7. 2D time history plots of the fire simulation outputs in PyroSim.

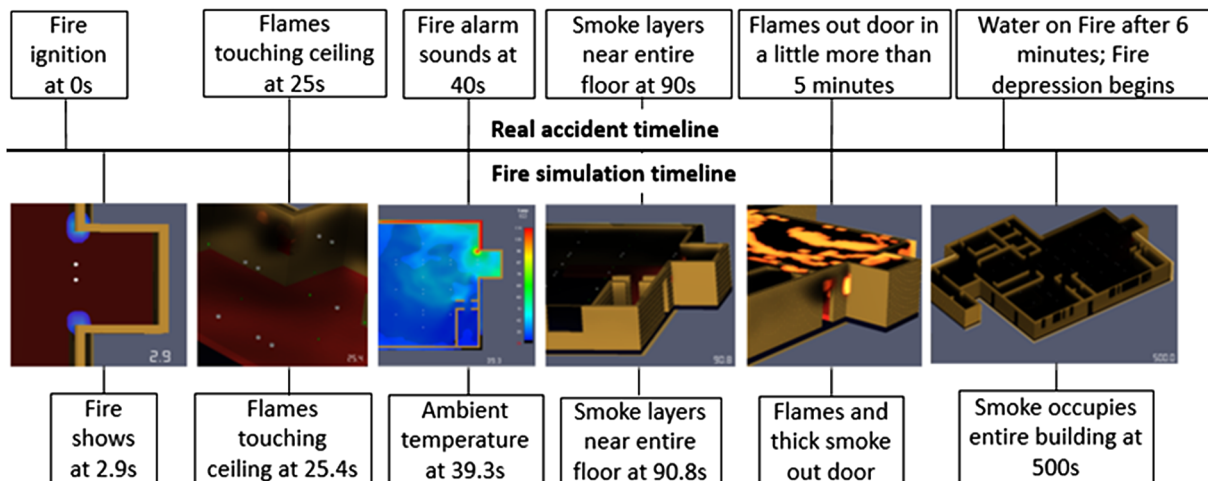


Fig. 6. The correspondence between two fire growth timelines.

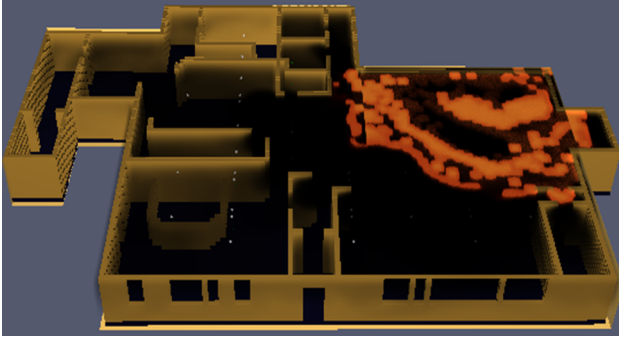


Fig. 8. Fire spreads fastest within the dance floor area in SMV.

times in order to avoid biased simulation outputs. As a result, the simulation results are summarized as: (1) 127 agents reached a shelter at 180 s, who ultimately escaped uninjured; (2) 215 agents reached a shelter between 180 and 300 s, who escaped with minor injuries; (3) 28 agents reached a shelter between 300 and 380 s, who escaped with more significant injuries; (4) 92 agents could not escape within 380 s and were considered dead; and (5) in the end, the average RSET for all the agents to reach shelter is calculated as 510 s. With regard to the numbers of injuries and casualties, Fig. 9 illustrates the correspondence between the accident records and our simulation results, which validates the accuracy of our simulation design. Besides, the pedestrian density flow indicates a fact—the primary entrance is frequently congested during the evacuation (Fig. 10)—as described in the accident report [56].

As can be seen, our experimental results of injuries/causalties are slightly fewer when compared to the real accident records. One of the reasons for this may be the fact that we did not have access to the actual as-built drawings, which would affect the fire simulation results in terms of ASETs, and further influence the number of agents at different ASETs. Nevertheless, our present design primarily concentrates on predicting evacuation performance. However, it was anticipated that the simulation would deliver a conservative prediction result. Due to the consideration of life safety, the conservativeness of evacuation simulation outcomes is important and should be studied in the future.

4.4. Statistical analysis

4.4.1. Building design optimization

According to [56], in addition to using highly flammable building materials, high occupant density and inadequate exit paths are primarily responsible for building fire disasters. To achieve both fire safety

and optimal building design goals, Eq. (1) is used to assess the evacuation performance for building design optimization while allowing time for all occupants to safely egress the facility. By fitting Eq. (1) in RStudio, the result indicates with a 95% confidence level that a very small p-value (< 0.01) significantly suggests a linear relationship between the RSET and different building designs in terms of number of exits, doorway width, and occupant capacity (Table 4). The estimated equation is provided below:

$$RSET = 265.60 - 32.78X - 46.78Y + 0.93Z \quad (2)$$

Using Eq. (2), one of the original building design variable values (number of exits, width of pathways, occupant capacity) can be changed, and compared to the original building design (Table 5). Consequently, the optimal solution was found to be that the original building design should be kept, but the occupant capacity should be lowered from 462 to 323.

4.4.2. Sensitivity analysis of the net heat flux

As it is difficult to predict the sizes of fires in the real world, this study conducted a sensitivity analysis to test the effect of the Net Heat Flux (NHF) on fire growth rate. The NHF, as defined as a critical factor when developing the FDS model, refers to the rate of heat energy transferred per surface unit area. The model parameter of NHF was adjusted to the target value in PyroSim prior to conducting the sensitivity analysis. Four levels of NHF (35, 40, 45, and 50 kW/m²) were used to test the time required to reach the following fire conditions: (1) the smoke layer height reaches 1.5 m and 1.2 m, (2) the temperature reaches 60 °C and 120 °C, and (3) the toxicity reaches 2500 ppm. As the histogram in Fig. 11 shows, there is a decreasing trend for the time required along with an increasing NHF and roughly decreasing vertical heights. Accordingly, the exponential curve, automatically generated with the histogram trend, can be interpreted as being indicative of an exponential growth rate between the fire growth and the NHF. Consequently, it coincides with the assumption made in Section 4.1, that is, the FDS model defines an exponential fire growth.

4.4.3. Evaluation of the designed agent-based evacuation system

With regard to the pedestrian movement design, the agents' evacuation performance is specified by six key model parameters, including receiving the fire signal, risk pending, group effect, decision altering system, counter-flow effect, and the effect of fire conditions. In particular, each model parameter is defined based on the critical factors summarized in Table 1. To evaluate the effectiveness of the model parameters, test scenarios were developed with and without the test parameters. As a result, when conducting two-sample *t*-tests, the very small p-values (< 0.01) verify the alternative hypothesis, while the 95% confidence levels of the mean differences are extracted to evaluate

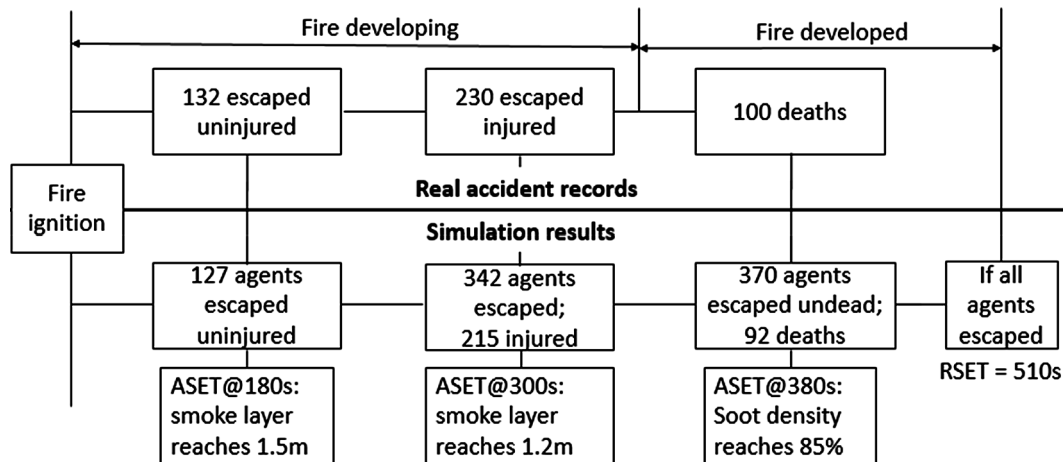


Fig. 9. The comparison of injuries and deaths.

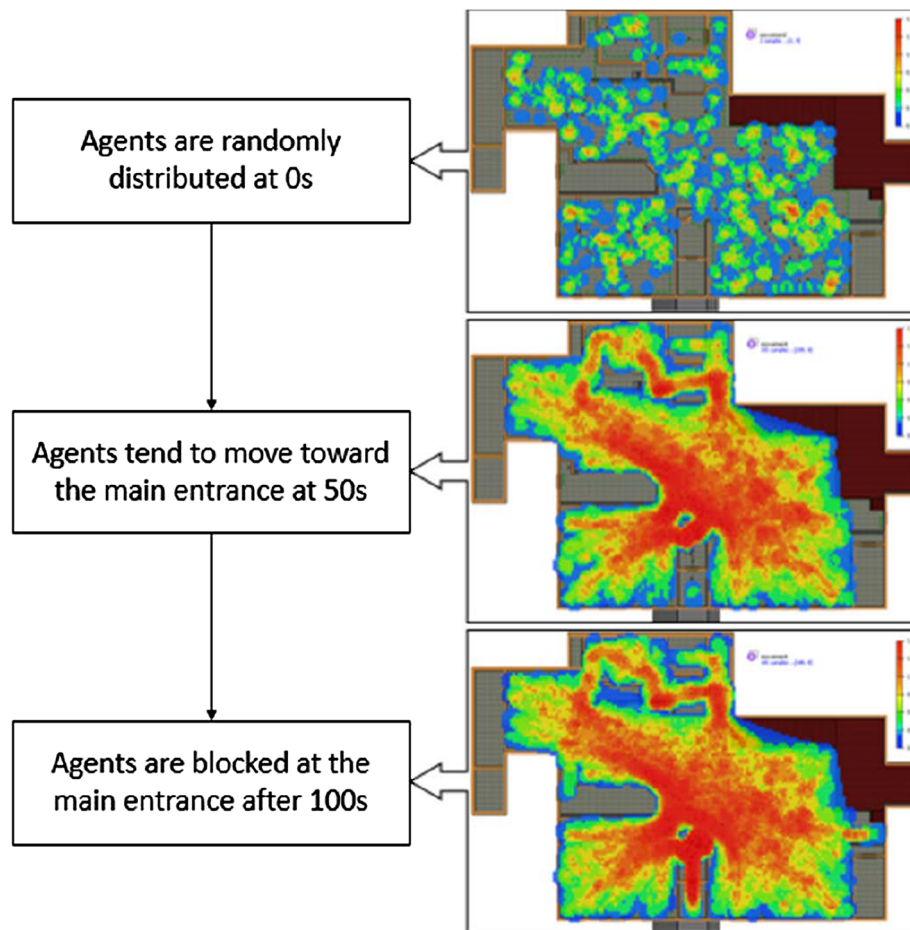


Fig. 10. The density flow of agents in Anylogic.

Table 4

Linear regression model outputs.

Explanatory variable	(Intercept)	Number of exits	Doorway width	Occupant capacity
Estimated coefficient	265.60	− 32.78	− 46.78	0.93
P-value	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Residual standard error: 30.39 on 23 df, F-statistic: 77.31 on 3 and 23 df, p-value: 3.664e-12				

Table 5

Alternative building designs for fire safety.

Categories	Number of exits (n)	Doorway width (m)	Occupant capacity (n)	RSET	ASET (for living)	RSET ≤ ASET?
Original design	4	1	462	517	380	No
Alternative#1	4	1	323	380	380	Yes
Alternative#2	8	1	462	378	380	Yes
Alternative#3	4	4	462	369	380	Yes

the model parameters' effects on the RSET value (Table 6). Consequently, the effectiveness of designed model parameters is validated, and their influences on evacuation performance are estimated.

Several findings can be drawn from Table 6: (1) the signal receiving stage increased the evacuation duration by 19.1–59.4 s due to the physical building properties, which include the thermal properties of materials, the location of fire cues, the sensitivity of the building's alarm system, etc.; (2) the risk pending stage increased the evacuation duration by 7.5–36.5 s due to agents' risk perception; (3) the guide effect decreased the evacuation duration by 28.1–64.3 s due to agent behaviors such as bounded rationality, familiarity and crowd attraction; (4) the re-decision system decreased the evacuation duration by

146.1–186.9 s due to agents' task fixation; (5) the counter-flow effect increased the evacuation duration by 8.1–36.9 s due to the negative effect of counter-flow, such as conjunctions; and (6) the effect of fire conditions increased the evacuation duration by 102.1–139.1 s due to the negative effects of temperature, smoke density and toxicity, and visibility and mobility of agents while evacuating.

In summary, the guide effect and re-decision system parameters positively impact the evacuation performance. Hence, to improve evacuation success, it is suggested to (1) increase the number of agents who are familiar with the safe exits in a given building to improve the guide effect and (2) provide adequate exit paths to avoid the phenomenon of task fixation. Moreover, to reduce the negative impacts of

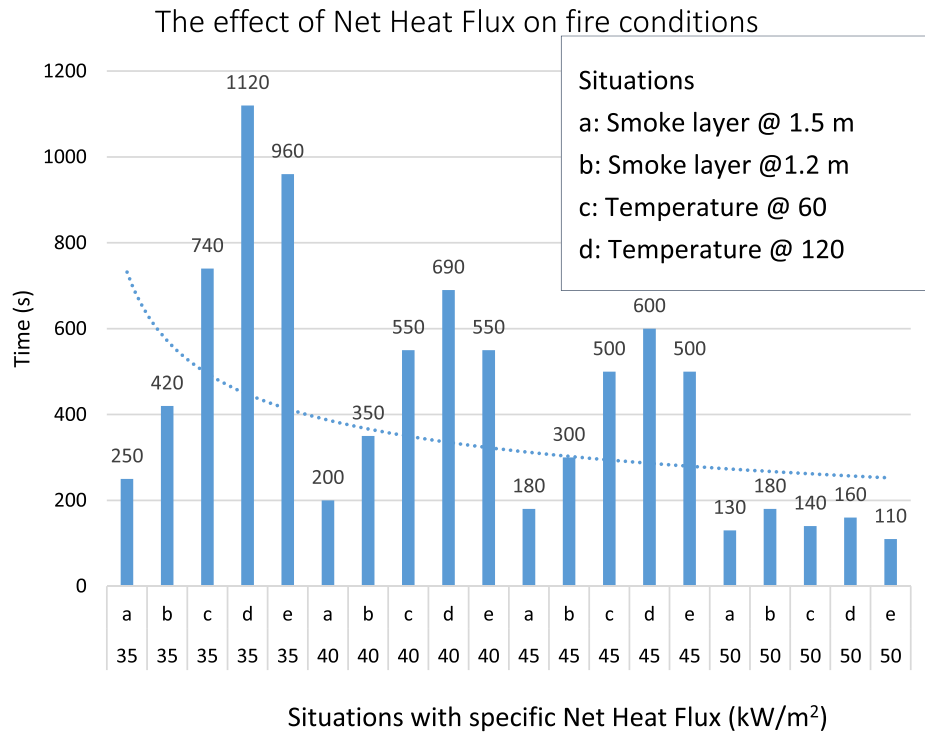


Fig. 11. The sensitivity analysis of fire conditions.

the other model parameters (i.e. the signal receiving stage, the risk pending stage, the counter-flow effect, and the effect of fire conditions), fire safety education can be a strategy to enhance fire safety awareness. Besides, optimized building layout design and inflammable materials used in buildings should be the standard practice.

4.5. Visualization through a 3D BIM environment

In terms of the simulation outputs, the dance floor area accounts for the fastest burn rate, as indicated by the SMV program in PyroSim. One possible explanation for this is that the building uses non-fire-retardant foam as wall insulation, while the nylon carpet speeds up the spread of flames. In AnyLogic, the pedestrian density flow congests the primary entrance during the evacuation, which may stem from both the group effect and pedestrians' familiarity with the exits' locations.

To learn from the simulation outputs, the dance floor area should be marked as a fire hazardous zone (Fig. 12), which hence needs to be fireproofed. Additionally, the side exits should be more clearly marked as recommended egress options as well as being displayed in the 3D walkthrough used for fire safety education (Fig. 13).

In addition, although it would be beneficial to visualize both fire and evacuation simulation outputs in a BIM platform simultaneously, this could not be achieved in this study due to technical limitations. However, future studies are recommended to take advantage of Revit

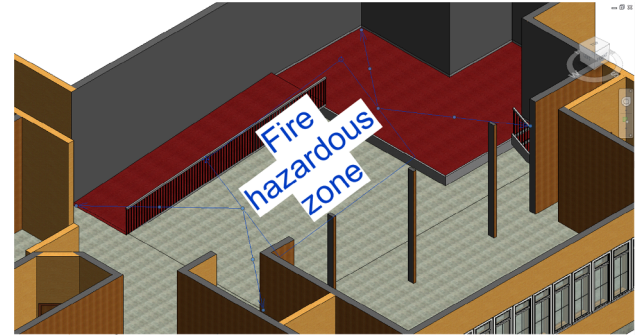


Fig. 12. The dance floor area is marked as fire hazardous zone in 3D Revit.

API, an application programming interface, to extend the capabilities of Revit for various simulation purposes.

5. Conclusions and future research

BIM can support the decision-making process regarding a building or other built asset. This study developed a BIM-based simulation framework that is capable of implementing FDS and ABM for fire safety management. To validate the proposed methodology, an experimental implementation involving the Station nightclub fire case is conducted.

Table 6

The outputs of two-sample *t*-tests.

Model scenarios	RSET mean	95% CI of mean difference	<i>t</i> -statistic	df	P-value
Original	509.4	–	–	–	–
Without the signal receiving stage	470.1	(19.2, 59.4)	4.1474	15.5	< 0.01
Without the risk pending stage	487.4	(7.5, 36.5)	3.1922	17.7	< 0.01
Without the guide effect	555.6	(– 64.3, – 28.1)	– 5.3808	16.7	< 0.01
Without the re-decision system	675.9	(– 186.9, – 146.1)	– 17.355	15.3	< 0.01
Without the counter-flow effect	486.9	(8.1, 36.9)	3.2838	17.7	< 0.01
Without the effect of fire conditions	388.8	(102.1, 139.1)	13.76	16.5	< 0.01

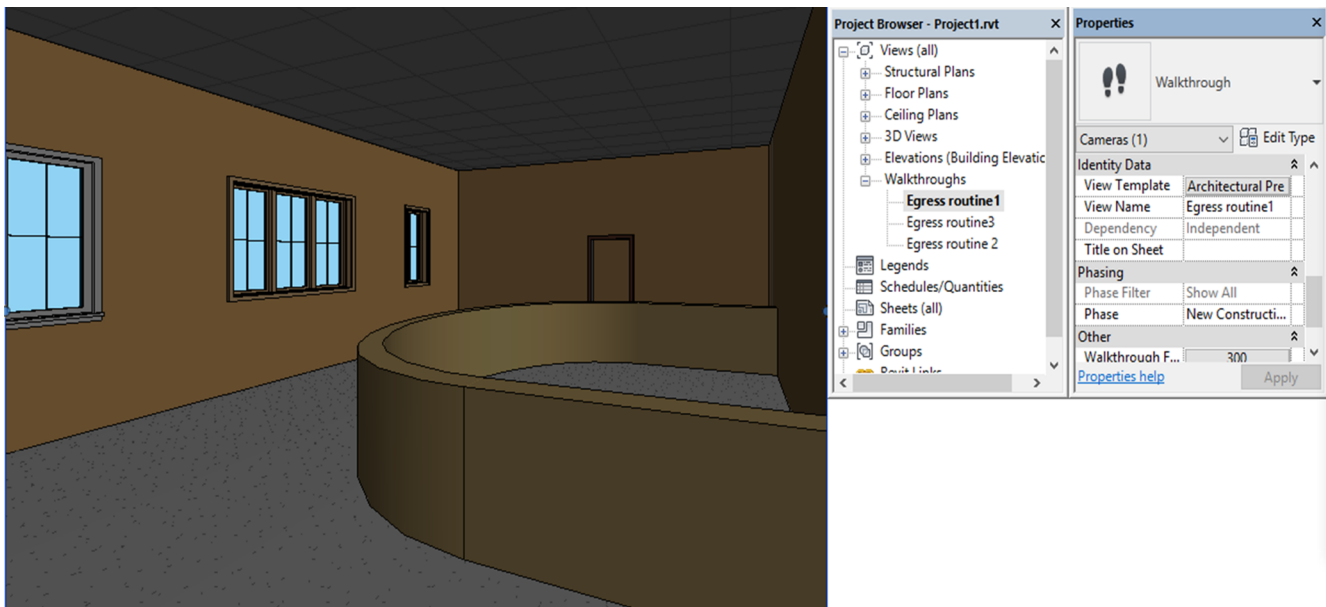


Fig. 13. Egress route displayed as a 3D walkthrough in Revit.

The analysis of the experimental results verifies the respective reliabilities of (1) using BIM as a platform for conducting simulation design, (2) simulating fire growth via the FDS technique, and (3) developing an ABM based evacuation scenario that accounts for the summarized critical factors (the physical properties of buildings, human behavior, and fire conditions). In particular, the simulation outputs are statistically analyzed by (1) fitting a linear regression model to investigate the relationship between the building design and the value of RSET, (2) verifying the exponential relationship between the fire growth rate and the NFH, and (3) applying two-sample *t*-tests to evaluate the effects of designed model parameters. Such an analysis helps to formulate better strategies for both fire safety design and fire safety education. Ultimately, 3D BIM serves as an environment for visualizing the results of (1) hazardous zones as reflected in the fire simulation and (2) effective escape routes as recommended by the evacuation scenario.

There are a few limitations to the current simulation framework design that should be noted. First, the experimental case study indicates fewer injuries and deaths when compared to the real accident records. However, it was anticipated that the simulation would deliver a more conservative prediction. In this study, the authors did not have access to as-built drawing of the studied building, which may have attributed to this discrepancy. Therefore, it can be concluded that it is important to have access to the actual as-built drawings, and other marginal effects (e.g. alcoholic beverages add fuel to the fire) should also be taken into consideration in order to improve the conservativeness of the fire simulation results in terms of ASETs so that all the agents can egress safely. Second, due to the limitations of the current FDS technique, this study sacrifices higher data resolution to shorten the computation time required to achieve an acceptable value. However, it is meaningful to achieve accurate simulation outputs without requiring excessive computation time. Third, the experimental implementation using a single-story building validated the practicability of the proposed framework. Though it can be said that a multi-story building could give additional insights for the evacuation performance. However, the researchers did not have access to the dataset of a well-documented multi-story building fire case to use in this study. Lastly, the simulation framework focuses on fire evacuation planning during the building design phase. However, it can be tailored for fire safety management during both the construction and maintenance phases.

To improve the practicability of the proposed framework, future studies are recommended to further (1) investigate human evacuation

behavior to improve the evacuation scenario design, (2) improve the computation performance of the FDS technique in terms of the data resolution, (3) implement the simulation framework for a multi-story building fire to foresee additional insights regarding the evacuation simulation performance, and (4) extend the BIM applications in relation to fire safety to the entire building life cycle.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] U.S. fire statistics: Trends in Fires, Deaths, Injuries and Dollar Loss. U.S. Fire Administration, <https://www.usfa.fema.gov/data/statistics/> (last accessed 2019/2/24).
- [2] W.N. Young, A.S. Jones, M.H. Bernstein, J.E. Gudgel, *The Business Value of BIM. SmartMarket Report: Design & Construction Intelligence*, McGraw-Hill Construction, 2009.
- [3] U. Rüppel, K. Schatz, Designing a BIM-Based Serious Game for Fire Safety Evacuation Simulations, *Adv. Eng. Inf.* 25 (4) (2011) 600–611.
- [4] B. Wang, H. Li, Y. Rezgui, A. Bradley, H. Ong, BIM based virtual environment for fire emergency evacuation, *Sci. World J.* 2014 (2014) 1–22.
- [5] J. Choi, J. Choi, I. Kim, Development of BIM-based evacuation regulation checking system for high-rise and complex buildings, *Autom. Constr.* 46 (2014) 38–49.
- [6] R. Eftekhariad, M. Nik-Bakht, A. Hammad, Extending IFC for fire emergency real-time management using sensors and occupant information, *EasyChair Preprints* (2018).
- [7] S. Wang, W. Wang, K. Wang, S. Shih, Applying building information modeling to support fire safety management, *Autom. Constr.* 59 (2015) 158–167.
- [8] NIST Website: FDS and Smokeview. <https://www.nist.gov/services-resources/software/fds-and-smokeview> (last accessed 2018/3/31).
- [9] T. Shen, Y. Huang, S. Chien, Using fire dynamic simulation (FDS) to reconstruct an arson fire scene, *Build. Environ.* 43 (6) (2008) 1036–1045.
- [10] E. Galea, Z. Wang, A. Veeraswamy, F. Jia, P. Lawrence, J. Ewer, Coupled fire/evacuation analysis of the station nightclub fire, *Fire Saf. Sci.* 9 (2008) 465–476.
- [11] J. Glasa, L. Valasek, P. Weisenpacher, L. Halada, Use of PyroSim for simulation of cinema fire, *Int. J. Recent Trends Eng. Technol.* 7 (2) (2012) 51–56.
- [12] S. Cao, W. Song, X. Liu, N. Mu, Simulation of pedestrian evacuation in a room under fire emergency, *Proc. Eng.* 71 (2014) 403–409.
- [13] EVAC. Software. AFMG Technologies GmbH.
- [14] F. Tingyong, X. Jun, Y. Jufen, W. Bangben, Study of building fire evacuation based on continuous model of FDS & EVAC, 2011 International Conference on Computer Distributed Control and Intelligent Environmental Monitoring, 2011.
- [15] E. Bonabeau, Agent-based modeling: methods and techniques for simulating human systems, *Proc. Natl. Acad. Sci.* 99 (Supplement 3) (2002) 7280–7287.

- [16] G. Bernardini, M. D'Orazio, E. Quagliarini, L. Spalazzi, An agent-based model for earthquake pedestrians' evacuation simulation in urban scenarios, *Transp. Res. Proc.* 2 (2014) 255–263.
- [17] G.P. Cimellaro, F. Ozzello, A. Vallero, S. Mahin, B. Shao, Simulating earthquake evacuation using human behavior models, *Earthquake Eng. Struct. Dyn.* 46 (6) (2017) 985–1002.
- [18] H. Wang, A. Mostafizi, L.A. Cramer, D. Cox, H. Park, An agent-based model of a multimodal near-field tsunami evacuation: decision-making and life safety, *Transport. Res. Part C: Emerg. Technol.* 64 (2016) 86–100.
- [19] B. Paveglio, T. Prato, Integrating Dynamic Social Systems into Assessments of future wildfire losses: an experiential agent-based Modeling Approach, Nova Science Publishers, 2011.
- [20] M.J. Widener, M.W. Horner, S.S. Metcalf, Simulating the effects of social networks on a population's hurricane evacuation participation, *J. Geogr. Syst.* 15 (2) (2012) 193–209.
- [21] Jumadi, S. Carver, D. Quincey, ABM and GIS-based multi-scenarios volcanic evacuation modelling of Merapi. *AIP Conference Proceedings* 1730, 050005, 2016.
- [22] S. Papadopoulos, E. Azar, Integrating building performance simulation in agent-based modeling using regression surrogate models: A novel human-in-the-loop energy modeling approach, *Energy Build.* 128 (2016) 214–223.
- [23] E. Azar, C.C. Menassa, A comprehensive analysis of the impact of occupancy parameters in energy simulation of office buildings, *Energy Build.* 55 (2016) 841–853.
- [24] E. Azar, C.C. Menassa, Agent-based modeling of occupants and their impact on energy use in commercial buildings, *J. Comput. Civil Eng.* 26 (4) (2012) 506–518.
- [25] F. Tang, A. Ren, Agent-based evacuation model incorporating fire scene and building geometry, *Tsinghua Sci. Technol.* 13 (5) (2008) 708–714.
- [26] J. Shi, A. Ren, C. Chen, Agent-based evacuation model of large public buildings under fire conditions, *Autom. Constr.* 18 (3) (2009) 338–347.
- [27] Y. Peizhong, W. Xin, L. Tao, Agent-based simulation of fire emergency evacuation with fire and human interaction model, *Saf. Sci.* 50 (4) (2011) 1171.
- [28] J. Joo, N. Kim, R. Wysk, L. Rothrock, Y. Son, Y. Oh, S. Lee, Agent-based simulation of affordable-based human behaviors in emergency evacuation, *Simul. Model. Pract. Theory* 32 (2013) 99–115.
- [29] P. Boguslawski, L. Mahdjoubi, V. Zverovich, F. Fadli, A dynamic approach for evacuees distribution and optimal routing in hazardous environments, *Autom. Constr.* 94 (2018) 11–21.
- [30] M. Kobes, I. Helsloot, B. Vries, J.G. Post, Building safety and human behavior in fire: a literature review, *Fire Saf.* 45 (1) (2010) 1–11.
- [31] Revit Software, Autodesk Inc.
- [32] Revit Website. Features. <https://www.autodesk.com/products/revit/features> (last accessed 2018/4/25).
- [33] C. Eastman, P. Teicholz, R. Sacks, K. Liston, *BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers and Contractors*, second ed., John Wiley & Sons, Inc., Hoboken, NJ, 2011.
- [34] PyroSim. Software. Thunderhead Engineering Consultants, Inc.
- [35] PyroSim Website. Features. <https://www.thunderheadeng.com/pyrosim/pyrosim-features/> (last accessed 2018/4/25).
- [36] T. Wang, M.H. Du, Y.F. Tang, Q. Zhang, An analysis on the fire model and the safety evacuation based on BIM, *Adv. Mater. Res.* 1065–1069 (2014) 2386–2389.
- [37] X. Chen, C. Liu, I. Wu, A BIM-based visualization and warning system for fire rescue, *Adv. Eng. Inf.* 37 (2018) 42–53.
- [38] Z. Xu, Z. Zhang, X. Lu, X. Zeng, H. Guan, Post-earthquake fire simulation considering overall seismic damage of sprinkler systems based on BIM and FEMA P-58, *Autom. Constr.* 90 (2018) 9–22.
- [39] Q. Sun, Y. Turkan, A BIM based simulation framework for fire evacuation planning, *Adv. Inform. Comput. Civ. Construct. Eng.* 431–438 (2018).
- [40] Netlogo. Software. Uri Wilensky.
- [41] Pathfinder. Software. Thunderhead Engineering Consultants, Inc.
- [42] Anylogic. Software. The AnyLogic Company.
- [43] N. Wagner, V. Agrawal, An agent-based simulation system for concert venue crowd evacuation modeling in the presence of a fire disaster, *Expert Syst. Appl.* 41 (6) (2014) 2807–2815.
- [44] R. Liu, D. Jiang, L. Shi, Agent-based simulation of alternative classroom evacuation scenarios, *Front. Arch. Res.* 5 (1) (2016) 111–125.
- [45] N. Mu, W. Song, X. Qi, W. Lu, S. Cao, Simulation of evacuation in a twin bore tunnel: analysis of evacuation time and egress selection, *Proc. Eng.* 71 (2014) 333–342.
- [46] Anylogic Website: Features. <https://www.anylogic.com/features/> (last accessed 2018/4/25).
- [47] M.T. Kinateter, E.D. Kuligowski, P.A. Reneke, R.D. Peacock, Risk perception in fire evacuation behavior revisited: definitions, related concepts, and empirical evidence, *Fire Sci. Rev.* 4 (1) (2015).
- [48] X. Pan, C.S. Han, K. Dauber, K.H. Law, A multi-agent based framework for the simulation of human and social behaviors during emergency evacuations, *AI & Soc.* 22 (2) (2007) 113–132.
- [49] S. Heliövaara, T. Korhonen, S. Hostikka, H. Ehtamo, Counterflow model for agent-based simulation of crowd dynamics, *Build. Environ.* 48 (2012) 89–100.
- [50] M. Zhong, C. Shi, X. Tu, T. Fu, L. He, Study of the human evacuation simulation of metro fire safety analysis in China, *J. Loss Prev. Process Ind.* 21 (3) (2008) 287–298.
- [51] K. Fridolf, K. André, D. Nilsson, H. Frantzich, The impact of smoke on walking speed, *Fire Mater.* 38 (7) (2013) 744–759.
- [52] P. Abolghasemzadeh, A comprehensive method for environmentally sensitive and behavioral microscopic egress analysis in case of fire in buildings, *Saf. Sci.* 59 (2013) 1–9.
- [53] AIA Document E202™-2013 Building Information Modeling Protocol. The American Institute of Architects. aiacontracts.org, 2013.
- [54] ASTM E84: Standard Test Method for Surface Burning Characteristics of Building Materials. ASTM Fire Standards. Six edition. American Society for Testing and Materials. West Conohocken, PA, 1000-1018, 2007.
- [55] Rstudio. Software. RStudio, Inc.
- [56] W.L. Grosshandler, N.P. Bryner, D. Madrzykowski, K. Kuntz, Report of the technical investigation of The Station nightclub fire. Gaithersburg, MD: U.S. Dept. of Commerce, Technology Administration, National Institute of Standards and Technology. NIST NCSTAR 2: Vol. I, 2005.